Finding the Inverse of a Matrix

1. Introduction to the Inverse of a Matrix

The **inverse of a matrix** is a matrix that gives the identity matrix when multiplied by the original matrix.

For a matrix A, its inverse is written as:

$$A^{-1}$$

The defining property is:

$$AA^{-1} = A^{-1}A = I$$

where:

- A = original matrix
- A^{-1} = inverse matrix
- I = identity matrix

The identity matrix acts like the number 1 in matrix multiplication.

Example of a 2×2 identity matrix:

$$I = \begin{bmatrix} 1 & 0 0 & 1 \end{bmatrix}$$

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2. Condition for the Existence of an Inverse

A matrix has an inverse only if its determinant is not zero.

The condition is:

$$\boxed{\det(A) \neq 0}$$

If:

$$\det(A) = 0$$

then:

- The matrix is **singular**.

- The inverse does not exist.

If:

$$\det(A) \neq 0$$

then:

- The matrix is **non-singular**.
- The inverse exists.

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3. Formula for Finding the Inverse

The inverse of a matrix using the adjugate method is:

$$\boxed{A^{-1} = \frac{1}{\det(A)}\,\mathrm{adj}(A)}$$

This means:

1. Find the determinant.
2. Find the adjugate.
3. Divide every element of the adjugate by the determinant.

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4. Steps to Find the Inverse of a Matrix

For a square matrix A:

Step 1:

Find the determinant:

$$\det(A)$$

Step 2:

Find the minors of all elements.

Step 3:

Convert minors into cofactors:

$$C_{ij} = (-1)^{i+j} M_{ij}$$

Step 4:

Form the cofactor matrix.

Step 5:

Transpose the cofactor matrix to get the adjugate:

$$\mathrm{adj}(A) = C^T$$

Step 6:

Use:

$$A^{-1} = \frac{1}{\det(A)} \mathrm{adj}(A)$$

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5. Inverse of a 2×2 Matrix

Consider:

$$A = \begin{bmatrix} a & b c & d \end{bmatrix}$$

Step 1: Find the determinant

$$\det(A) = ad - bc$$

Step 2: Find the adjugate

For a 2×2 matrix:

$$\mathrm{adj}(A) = \begin{bmatrix} d & -b - c & a \end{bmatrix}$$

Step 3: Apply the inverse formula

$$\boxed{A^{-1} ====== \frac{1}{ad - bc} \left[d \quad -b \; - c \quad a \right]}$$

6. Example: Finding the Inverse of a 2×2 Matrix

Find the inverse of:

$$A = \left[2 \quad 3\,4 \quad 5 \right]$$

Step 1: Find the determinant

Using:

$$\det(A) = ad - bc$$

Substitute values:

$$\det(A) = (2)(5) - (3)(4)$$

$$= 10 - 12$$

Therefore:

$$\det(A) = -2$$

Since:

$$-2 \neq 0$$

the inverse exists.

Step 2: Find the adjugate

For:

$$A = \begin{bmatrix} 2 & 3 4 & 5 \end{bmatrix}$$

Swap the diagonal elements:

$$2 \leftrightarrow 5$$

Change signs of the other elements:

$$3 \to -3$$

$$4 \to -4$$

Therefore:

$$\mathrm{adj}(A) ====================== \begin{bmatrix} 5 & -3 - 4 & 2 \end{bmatrix}$$

Step 3: Apply the formula

$$A^{-1} ====== \frac{1}{-2} \begin{bmatrix} 5 & -3 - 4 & 2 \end{bmatrix}$$

Multiply each element by $-\frac{1}{2}$:

$$\boxed{A^{-1} ====== \begin{bmatrix} -\frac{5}{2} & \frac{3}{2} 2 & -1 \end{bmatrix}}$$

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7. Verification of the Inverse

To check the answer:

$$AA^{-1} = I$$

Multiply:

$$\begin{bmatrix} 2 & 3 4 & 5 \end{bmatrix} \begin{bmatrix} -\frac{5}{2} & \frac{3}{2} 2 & -1 \end{bmatrix}$$

First element:

$$(2)(-\frac{5}{2}) + (3)(2)$$

$$= -5 + 6 = 1$$

Second element:

$$(2)(\frac{3}{2}) + (3)(-1)$$

$$= 3 - 3 = 0$$

Third element:

$$(4)(-\frac{5}{2}) + (5)(2)$$

$$= -10 + 10 = 0$$

Fourth element:

$$(4)(\frac{3}{2}) + (5)(-1)$$

$$= 6 - 5 = 1$$

Therefore:

$$AA^{-1} ======= \begin{bmatrix}1 & 0\,0 & 1\end{bmatrix}$$

The inverse is correct.

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8. Finding the Inverse of a 3×3 Matrix

For:

$$A = \begin{bmatrix}a & b & c\,d & e & f\,g & h & i\end{bmatrix}$$

The process is longer because we need nine cofactors.

Step 1: Find the determinant

Expand using cofactors:

$$\det(A) ======= aC_{11} + bC_{12} + cC_{13}$$

or:

$$\det(A) ======= a(ei - fh) - b(di - fg) + c(dh - eg)$$

Step 2: Find all cofactors

Calculate:

$$C_{11}, C_{12}, ..., C_{33}$$

The cofactor matrix is:

$$C = \left[C_{11} \quad C_{12} \quad C_{13}\, C_{21} \quad C_{22} \quad C_{23}\, C_{31} \quad C_{32} \quad C_{33} \right]$$

Step 3: Find the adjugate

Transpose the cofactor matrix:

$$\mathrm{adj}(A) = C^T$$

Step 4: Apply the formula

$$\boxed{A^{-1} ====== \frac{1}{\det(A)} \mathrm{adj}(A)}$$

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9. Example: Inverse of a 3×3 Matrix

Find the inverse of:

$$A = \left[1 \quad 2 \quad 3\, 0 \quad 4 \quad 5\, 1 \quad 0 \quad 6 \right]$$

Step 1: Find determinant

Expand along the first row:

$$\det(A) ======= 1 \left| 4 \quad 5\, 0 \quad 6 \right| -------------- 2 \left| 0 \quad 5\, 1 \quad 6 \right| + 3 \left| 0 \quad 4 \right.$$

Calculate each part:

First minor:

$$(4)(6) - (5)(0) = 24$$

Second minor:

$$(0)(6) - (5)(1) = -5$$

Third minor:

$$(0)(0) - (4)(1) = -4$$

Substitute:

$$\det(A) = 1(24) - 2(-5) + 3(-4)$$

$$= 24 + 10 - 12$$

Therefore:

$$\det(A) = 22$$

Since:

$$22 \neq 0$$

the inverse exists.

Step 2: Find the Cofactor Matrix

After calculating all cofactors:

$$C = \begin{bmatrix} 24 & 5 & -4 - 12 & 3 & 2 - 2 & -5 & 4 \end{bmatrix}$$

Step 3: Find the Adjugate

Transpose:

$$\text{adj}(A) = C^T$$

Therefore:

$$\text{adj}(A) = \begin{bmatrix} 24 & -12 & -2\,5 & 3 & -5 - 4 & 2 & 4 \end{bmatrix}$$

Step 4: Find the Inverse

Use:

$$A^{-1} ====== \frac{1}{22} \left[24 \quad -12 \quad -2\; 5 \quad 3 \quad -5 - 4 \quad 2 \quad 4 \right]$$

Therefore:

$$\boxed{A^{-1} ====== \left[\frac{24}{22} \quad -\frac{12}{22} \quad -\frac{2}{22}\; \frac{5}{22} \quad \frac{3}{22} \quad -\frac{5}{22} - \frac{4}{22} \quad \frac{2}{22} \quad \frac{4}{22} \right]}$$

10. Important Properties of Matrix Inverse

1. Inverse of an inverse

$$(A^{-1})^{-1} = A$$

2. Inverse of a product

$$(AB)^{-1} = B^{-1}A^{-1}$$

The order reverses.

3. Inverse of transpose

$$(A^T)^{-1} = (A^{-1})^T$$

4. Identity matrix

$$I^{-1} = I$$

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11. Singular and Non-Singular Matrices

Singular Matrix

A matrix is singular if:

$$\det(A) = 0$$

Properties:

- No inverse exists.
- Rows or columns are dependent.

Example:

$$A = \begin{bmatrix} 1 & 2\,2 & 4 \end{bmatrix}$$

Determinant:

$$(1)(4) - (2)(2) = 4 - 4 = 0$$

Therefore:

$$A^{-1} \text{ does not exist}$$

Non-Singular Matrix

A matrix is non-singular if:

$$\det(A) \neq 0$$

It has a unique inverse.

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12. Applications of Matrix Inverse

Matrix inverses are used in:

- Solving simultaneous equations
- Computer graphics transformations

- Engineering calculations
- Cryptography
- Machine learning
- Physics simulations

For solving equations:

$$AX = B$$

we multiply both sides by A^{-1}:

$$A^{-1}AX = A^{-1}B$$

Since:

$$A^{-1}A = I$$

we get:

$$\boxed{X = A^{-1}B}$$

Final Summary

Step	Operation
1	Calculate $\det(A)$
2	Find minors
3	Convert minors into cofactors
4	Form cofactor matrix
5	Transpose to get $\operatorname{adj}(A)$
6	Use $A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$

The complete chain is:

$$\boxed{\text{Determinant} \rightarrow \text{Minors} \rightarrow \text{Cofactors} \rightarrow \text{Adjugate} \rightarrow \text{Inverse}}$$